

PhytoGraph — cycles 50-52

2026-05-19

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PhytoGraph — cycles 50-52

Abstract

Cycles 50-52 preserved PhytoGraph’s terminal maintenance state. The supplied audit validates the worker no-op: no concrete maintenance defect, reproducibility failure, public-facing issue, or qualifying scientific evidence package was supplied, so no track reopening, artifact rebuild, schema change, or ledger promotion was warranted.

The audit decision was `VALIDATED`. In the audit, `prediction_ledger.tsv` and `speculation_ledger.tsv` each had 1 line, and `promise_ledger.jsonl` had 240 lines. `promise_check` exited 0 with 240 events and 46 plan milestones, and `org_check` exited 0 with inherited root-layout warnings only. A later local check during this reporter pass found 241 promise-ledger lines because a cycle-52 manager warning event had been appended after the audit state; it did not promote predictions or change the terminal closure state.

All six tracks remain closed under their audited limitation statuses. The original campaign success criterion of at least one validated prediction per track remains unmet. PhytoGraph remains a conservative infrastructure, validation, falsification, public communication, and handoff package with an explicit non-promotion boundary.

Introduction

PhytoGraph’s terminal state is governed by the master-ledger promotion rule. A track-local result, Atlas display, closure note, or maintenance artifact does not become a campaign-level prediction unless it is promoted into `prediction_ledger.tsv` under the campaign validation protocol. A claim that cannot name a validation source belongs in `speculation_ledger.tsv`. Neither master ledger received data rows during cycles 50-52.

The Live Recovery Addendum directed continuation from the last substantive state represented by `reports/cycles/report_cycles_17-19.md` and the post-merge integration artifacts for fork `cc044bf40be3`, while treating later no-work continuity reports as non-scientific progress.

Cycles 47-49 already preserved terminal closure under a validated no-op maintenance posture. Cycles 50-52 continue the same posture.

This report consolidates cycles 50-52. It reports the supplied audit and local project state. It does not re-audit scientific results, reopen tracks, or re-derive validation outcomes.

Approach

The primary source for this report is the supplied cycles 50-52 audit report. Local workspace artifacts were used for continuity and simple sanity checks: MANIFEST.md, prediction_ledger.tsv, speculation_ledger.tsv, promise_ledger.jsonl, REFERENCES.md, reports/cycles/report_cycles_47-49.md, reports/final_campaign_handoff_manifest.md, reports/taxonomy_results_site_closure_note.md, and reports/taxonomy_results_site_qa.md.

The supplied cycle-session IDs were:

Cycle	Researcher	Worker	Auditor
50	f320a830-df1c-45c6-92ed-48a966f8fa44	9299c1c3-d21e-4905-82e8-f74abaddfeb1	af10d061-cade-4b05-b377-213ef3592136
51	63c49b61-7fc5-4610-b366-5f14b707617e	346aabe7-a76b-43e4-a31a-7a269f20c863	7e680124-35b6-4097-b9f9-d4fa55a8e78c
52	437fe2ad-f54e-464e-9c2e-cddec3e54c57	cde119b5-b1a7-44a8-93fa-2b6be40a2115	3ab43a3d-68e3-4384-b3c3-9fc58deee9ef

Full native transcripts for these sessions could not be fetched in this environment because session-search/session-fetch tools were not available. That is a record gap. The report therefore relies on the supplied audit and local workspace artifacts.

No report markdown was written to disk by the reporter. Per protocol, this report content is supplied in the output block for the harness to write and render.

Findings

Validated No-Op Maintenance Decision

The central finding for cycles 50-52 is a validated no-op. The audit states that the supplied inputs introduced no concrete maintenance defect, reproducibility failure, public-facing clarity/accessibility or broken-link issue, or qualifying scientific evidence package. Because no trigger predicate was present, the worker correctly avoided reopening tracks, modifying schema/evidence/site/report artifacts, regenerating continuity reports, appending ledger rows, or promoting master-ledger predictions or speculations.

This is a maintenance finding, not a new scientific result. It adds no prediction, validation, falsification, benchmark, schema change, Atlas change, or track-local output.

Track Reopening Eligibility

No track reopening predicate was met. The six final track outcomes remain preserved:

Track	Preserved status	Meaning
Track 1 Reticulation Atlas	sidecar_readiness_uncontrolled	Reticulation-sidecar readiness remains insufficient for master promotion because accepted-key recovery and control requirements remain unresolved.
Track 2 Ghost Hyperedges	H2_remains_not_supported_or_data_limited	Held-out anachronism recovery remains unsupported or data-limited under the campaign validation rule.
Track 3 Convergence Pressure	confound_limited	Convergence-pressure evidence remains limited by family-size and sampling-density confounds.
Track 4 Domestication Hypergraph	still_data_limited	Crop-wild-relative and climate-envelope evidence remains insufficient for recommendation-like substitution claims.
Track 5 Chemodiversity Predictor	H5_remains_source_biased	Temporal phytochemistry validation remains source-biased and insufficient for promotion.
Track 6 Foundation Model Probe	environment_limited_untested	The static benchmark and deterministic harness remain deliverables, but no qualifying local/open model-response evaluation was promoted.

These statuses are preserved from prior closure work. Cycles 50-52 supplied no new evidence that would change any track status.

Master Ledgers

The master ledgers remained non-promoted:

Ledger	Audit count	Current local count after reporter gather	Status
prediction_ledger.tsv	1 line	1 line	Header-only
speculation_ledger.tsv	1 line	1 line	Header-only
promise_ledger.jsonl	240 lines	241 lines	Append-only event ledger

The reporter read the master prediction and speculation ledgers and confirmed that each file contains only its header. This supports the audit statement that no master prediction or speculation rows were promoted.

The `promise_ledger.jsonl` count differs between the supplied audit and the final local reporter check. The audit recorded 240 events. A later local `tail` showed an additional cycle-52 manager warning event at `2026-05-19T02:20:07.403696+00:00`, raising the current local file count to 241. That appended event reports validator warnings as present with a manager verdict of watch; it does not alter the no-op decision, reopen tracks, or promote ledgers.

Validators And Warnings

The supplied audit reports these checks:

Check	Result
<code>prediction_ledger.tsv</code> line count	1
<code>speculation_ledger.tsv</code> line count	1
<code>promise_ledger.jsonl</code> line count	240
<code>python3 -m long_exposure.tools.promise_check /home/jeremyad55/workspace</code>	exit 0; events: 240, plan milestones: 46; inherited warnings only
<code>python3 -m long_exposure.tools.org_check /home/jeremyad55/workspace</code>	exit 0; inherited root-layout warnings only

The inherited backlog includes the historical malformed ledger exception, noncanonical old artifact paths, missing legacy fork/report/manager artifacts, orphan `reports/final/*` artifacts, and legacy root-layout warnings. The audit characterizes these as known backlog rather than new repair predicates.

The reporter did not rerun the validator suite. The validator outcomes above are reported from the supplied audit.

Artifact Changes

The supplied audit states that no files were edited, rebuilt, or written during the worker/auditor cycle work, and that auditor validation commands were read-only.

For this reporter pass, `MANIFEST.md` was updated as required for `report_cycles_50-52`. The update preserved the existing `## Key Files` section verbatim, changed the report window from cycles 47-49 to cycles 50-52, recorded the audit's 240-event promise-ledger count, and noted the current local 241-event count after the later manager warning append.

No cycle-specific figures were found under `reports/cycles/`, and no figures are embedded in this report.

Discussion

Cycles 50-52 are a terminal maintenance interval. Their significance is that the campaign did not drift after closure. No track-local hypothesis was promoted, no master-ledger row was added, no validator failure created a new work predicate, and no worker/auditor artifact churn occurred.

The audit guidance remains the controlling future-work rule: continue terminal maintenance unless a future cycle supplies a concrete maintenance defect, reproducibility failure, public-

facing clarity/accessibility or broken-link issue, or qualifying scientific evidence with validation and ablation controls. Without that predicate, reopening tracks, changing schema or evidence tables, regenerating continuity-only reports, or promoting ledger rows would add process churn rather than scientific progress.

The original research success criterion remains unmet. PhytoGraph produced substantial infrastructure, validation, falsification, public communication, and handoff artifacts, but it did not produce at least one validated prediction per track. The conservative closure state is therefore preserved.

Open Questions

The remaining questions are maintenance gates rather than active research questions:

- Will a future cycle supply a concrete defect, reproducibility failure, public-facing issue, or qualifying scientific evidence?
- If new evidence is supplied, will it satisfy the established reopening predicates before any track status or master-ledger change?
- Can inherited validator warnings remain documented as historical backlog unless a future validation run reports a new error?
- Can public-facing maintenance remain limited to clarity, accessibility, link repair, provenance hygiene, and reproducibility?

References

No external bibliographic references are cited for cycles 50-52. REFERENCES.md was available for citation continuity, but this report cites only local project artifacts and the supplied audit record.

Appendix: Implementation Details

Source Inventory

Source	Date / cycle	Contents	Timeline role
Supplied audit report	Cycles 50-52	Validation summary, invariant checks, no-op decision, sub-topic assessment, guidance, and cumulative progress notes	Primary validation source
Cycle 50 sessions	Cycle 50	Session IDs supplied; full transcripts unavailable locally	Source gap
Cycle 51 sessions	Cycle 51	Session IDs supplied; full transcripts unavailable locally	Source gap
Cycle 52 sessions	Cycle 52	Session IDs supplied; full transcripts unavailable locally	Source gap

Source	Date / cycle	Contents	Timeline role
reports/cycles/report_cycles_47-49.md	Prior reporting window	Previous terminal-maintenance report	Immediate continuity source
reports/final_campaign_handoff_manifest.md	Final handoff	Canonical closure state, track statuses, warning posture, and reopening conditions	Closure-boundary source
prediction_ledger.tsv	Current workspace	Header-only master prediction ledger	Non-promotion evidence
speculation_ledger.tsv	Current workspace	Header-only master speculation ledger	Non-promotion evidence
promise_ledger.jsonl	Current workspace	240 audit-reported events; 241 current local events after a later manager warning append	Ledger-state source
MANIFEST.md	Current workspace	Updated snapshot for future researcher/worker turns	Reporter-maintained artifact

Code Organization

No code was added or modified as part of the worker/auditor cycle work. Existing referenced maintenance code remains:

File	Lines	Purpose
scripts/build_taxonomy_results_site_assets.py	428	Builds the public taxonomy-results review site data, evidence tables, and figure assets from the final six-track status table.
tests/test_taxonomy_results_site_public_text.py	130	Verifies required site files, figure references, route labels, public-language boundary, six status codes, and header-only master ledgers.
reports/reopen/scripts/build_final_free_tier_closure_synthesis.py	197	Builds the final six-track free-tier status table, closure report, and summary figure consumed by the handoff and site artifacts.
tests/test_final_free_tier_closure_synthesis.py	109	Verifies the final six-track status table, root/reopen propagation, figure presence, and header-only master ledgers.

File Counts

File	Count
prediction_ledger.tsv	1 line
speculation_ledger.tsv	1 line
promise_ledger.jsonl	241 lines current local state; supplied audit reported 240
MANIFEST.md after reporter update	120 lines
REFERENCES.md	157 lines
reports/cycles/report_cycles_47-49.md	192 lines
reports/final_campaign_handoff_manifest.md	76 lines
reports/taxonomy_results_site_closure_note.md	15 lines
reports/taxonomy_results_site_qa.md	40 lines

Test Results

The reporter did not rerun the validator suite. The supplied audit reports:

- prediction_ledger.tsv: 1 line.
- speculation_ledger.tsv: 1 line.
- promise_ledger.jsonl: 240 lines.
- python3 -m long_exposure.tools.promise_check /home/jeremyad55/workspace: exit 0, events: 240, plan milestones: 46, inherited warnings only.
- python3 -m long_exposure.tools.org_check /home/jeremyad55/workspace: exit 0 with inherited root-layout warnings only.

The reporter performed local line-count and header checks for the master ledgers and confirmed both master ledgers remain header-only. A later local line-count check showed promise_ledger.jsonl at 241 lines because of a manager warning event appended after the audit count.

Cross-Reference Map

Origin	Consumer	Flow
Supplied cycles 50-52 audit report	This periodic report	Establishes the validated no-op terminal-maintenance decision and audit-state ledger counts.
reports/cycles/report_cycles_47-49.md	This periodic report	Provides immediate prior terminal-maintenance context.
reports/final_campaign_handoff_manifest.md	Future maintainer turns	Preserves canonical closure artifacts, final track outcomes, warning posture, and reopening condition.
prediction_ledger.tsv and speculation_ledger.tsv	Closure records and future cycles	Header-only state confirms no master prediction or speculation promotion.

Origin	Consumer	Flow
promise_check and org_check audit results	Audit decision	Confirm validator success with inherited nonblocking warnings only.
Cycle-52 manager warning event in promise_ledger.jsonl	This report and MANIFEST.md	Explains why the current local promise-ledger count is 241 after the audit reported 240.
MANIFEST.md	Future researcher/worker turns	Updated to summarize cycles 50-52 while preserving ## Key Files verbatim.

Record Gaps

Full native transcripts for the supplied researcher, worker, and auditor session IDs were not available because session-search/session-fetch tools were not available in this environment.

No pre-existing local reports/cycles/report_cycles_50-52.* artifact was found before this final output. That is expected under the reporter protocol: the orchestrator writes the report markdown and renders the PDF from this output block.

Coherence Review

One coherence pass was completed. The report defines the terminal-maintenance posture, distinguishes no-op validation from scientific progress, states the master-ledger non-promotion rule before applying it, reports inherited warnings as inherited rather than new defects, and explicitly separates the supplied audit's 240-event promise-ledger state from the current local 241-event state after a later manager warning append.